

181755/171755/

121755/031755

Sub : CNC Machines & Automation

M.M. : 100

- Q.6 A robot's arm is also known as its (CO7)
a) Actuator b) End effector
c) Manipulator d) Servomechanism
- Q.7 ASRS means (CO6)
a) Auto start rest stop
b) Automatic storage and retrieval system
c) Auto sensor and range sensor
d) None of the above
- Q.8 Problems in electrical components are: (CO5)
a) Cylinder b) Piston
c) Valve d) Solenoids
- Q.9 _____ is not a type of NC program reader: (CO1)
a) Punched tape reader
b) Laser tape reader
c) Card reader
d) Mechanical tape reader
- Q.10 Which of the following are not components of flexible tooling: (CO7)
a) Tool magazine
b) Tool changing mechanism
c) Encoder d) Work holder

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 _____ machine can control many machines at same time. (CO1)

- Q.12 CNC machine _____ (increase / decrease) the lead time. (CO3)
- Q.13 Expand MCU. (CO1)
- Q.14 A motor used in control with feedback is _____. (CO6)
- Q.15 Write the Function of Tachometer. (CO6)
- Q.16 G90 stands for _____. (CO2)
- Q.17 _____ code is used for DWELL function. (CO2)
- Q.18 Write full form of AGV. (CO6)
- Q.19 Define CIM. (CO7)
- Q.20 Define end effector. (CO7)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

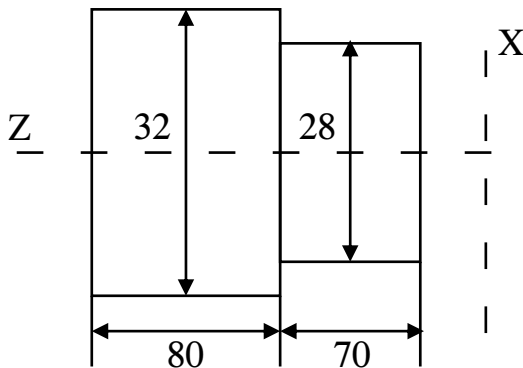
- Q.21 Define NC, CNC and DNC system. (CO1)
- Q.22 Explain the working of machine control unit. (CO1)
- Q.23 Write the advantages of re-circulating ball screws with conventional lead screws. (CO2)
- Q.24 Discuss any two swarf removal methods in CNC machine. (CO2)
- Q.25 Write short note on slide ways. (CO1)
- Q.26 Define actuators. Explain Pneumatic actuators. (CO6)
- Q.27 Write short note of opto-interrupter. (CO1)
- Q.28 Explain active and passive transducers. (CO6)
- Q.29 Distinguish between G-Codes and M-Codes. (CO3)

- Q.30 Write the differences between fixed zero and floating zero. (CO3)
- Q.31 Explain with examples the operation of a canned cycle. (CO4)
- Q.32 Write the common problems in mechanical parts of a CNC. (CO5)
- Q.33 Explain any two fault finding techniques. (CO5)
- Q.34 Differentiate between point to point & continuous path motion of a robot. (CO7)
- Q.35 Write short note on FMS. (CO6)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Define DNC system. Explain its working. Write the advantages and disadvantages of DNC system. (CO1)
- Q.37 Explain different types of automation along with advantages and limitations. (CO6)
- Q.38 Write a simple part program for step turning. (CO4)



(1220)

(4)

181755/171755/
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